=== UNIT 5: DYNAMICS OF PARTICLES: KINEMATICS & KINETICS ===

>> 5.1 Rectilinear Kinematics of Particles

Dynamics is the branch of mechanics dealing with accelerated bodies, subdivided into Kinematics (geometry of motion without reference to forces) and Kinetics (relating forces to resulting accelerations and masses).
* Fundamental Differential Relations: Displacement s(t), Velocity v(t) = ds/dt, Acceleration a(t) = dv/dt = d^2s/dt^2 = v * (dv/ds).
* Constant Acceleration Equations (a = const):
v = u + a*t.
s = u*t + 0.5*a*t^2.
v^2 = u^2 + 2*a*s.
* Variable Acceleration: Requires direct analytical integration:
- a = f(t): v = Integral(f(t) dt); s = Integral(v dt).
- a = f(v): t = Integral(dv / f(v)); s = Integral(v dv / f(v)).
- a = f(s): v^2 = 2 * Integral(f(s) ds) + C.

>> 5.2 Curvilinear Motion: Normal and Tangential Coordinates

When a particle travels along a curved trajectory, its acceleration vector is naturally decoupled into orthogonal components along the tangent (t) and principal normal (n) to the path:
* Tangential Velocity: v = ds/dt (always directed tangentially to the curve).
* Tangential Acceleration: a_t = dv/dt = v * (dv/ds). Represents rate of change of speed (magnitude).
* Normal (Centripetal) Acceleration: a_n = v^2 / rho, where rho is local radius of curvature: rho = [1 + (dy/dx)^2]^(3/2) / |d^2y/dx^2|. Represents rate of change of direction (always directed inward toward the center of curvature).
* Total Acceleration Magnitude: a = sqrt(a_t^2 + a_n^2). Direction angle: phi = atan(a_n / a_t).

>> 5.3 Kinetics of Particles: Newton’s Second Law & D’Alembert’s Principle

* Newton’s Second Law: Sigma F = m * a. In Cartesian components: Sigma Fx = m * a_x; Sigma Fy = m * a_y. In path coordinates: Sigma F_t = m * a_t; Sigma F_n = m * (v^2 / rho).
* D’Alembert’s Principle: Converts a dynamic acceleration problem into an equivalent static equilibrium problem by introducing a fictitious Inertia Force: F_inertia = - m * a.
Dynamic Equilibrium Equation: Sigma F + (- m * a) = 0.
Engineers can then apply standard equations of static equilibrium (Sigma Fx = 0, Sigma

Fy = 0) including the reversed inertia force.

>> 5.4 Work-Energy Principle for Particles

* Work of a Force: U_1-2 = Integral(F * cos(theta) ds) = Integral(F dot dr).
- Constant Force: U = F * s * cos(theta).
- Gravity Force (Weight): U = - W * Delta h (positive if moving down, negative if moving up).
- Linear Elastic Spring Force: U = - 0.5 * k * (x2^2 - x1^2).
* Kinetic Energy: KE = 0.5 * m * v^2.
* Work-Energy Theorem: Total work done by all external forces acting on a particle equals the change in its kinetic energy: Sigma U_1-2 = KE_2 - KE_1 = 0.5*m*v2^2 - 0.5*m*v1^2.
* Conservation of Mechanical Energy: For conservative force fields (gravity, ideal springs): KE_1 + PE_1 = KE_2 + PE_2 = constant.

>> 5.5 Impulse-Momentum Principle

* Linear Momentum: p = m * v (vector quantity in kg-m/s or N-s).
* Linear Impulse: J = Integral_t1^t2(F dt).
* Principle of Linear Impulse and Momentum: The impulse of resultant force equals the change in linear momentum: Integral_t1^t2(Sigma F dt) = m*v2 - m*v1.
* Conservation of Linear Momentum: If resultant external force acting on a system of particles vanishes along an axis (Sigma F_ext = 0), total linear momentum along that axis is conserved: Sigma(m_i * v_i1) = Sigma(m_i * v_i2).

>> 5.6 Impact and Collision Mechanics

* Direct Central Impact: Collinear collision of two bodies where velocities are directed along the line of impact passing through their centers of mass.
* Coefficient of Restitution (e): The ratio of relative velocity of separation after impact to relative velocity of approach before impact:
e = (v_B2 - v_A2) / (v_A1 - v_B1) = Relative Velocity of Separation / Relative Velocity of Approach.
* Collision Regimes: (1) Perfectly Elastic Impact: e = 1.0 (no loss of kinetic energy). (2) Inelastic (Plastic) Impact: 0 < e < 1.0 (kinetic energy lost to heat and acoustic deformation). (3) Perfectly Plastic Impact: e = 0 (bodies coalesce and move together with common velocity: v_A2 = v_B2).

[EXAMPLE 1: Kinematics of Particle with Variable Acceleration]

Problem Context: A civil engineering piling hammer moves along a vertical guide with variable acceleration given by a(t) = (12*t - 3*t^2) m/s^2, where t is in seconds. At t = 0, displacement s(0) = 0 m and initial downward velocity v(0) = 4.0 m/s. Calculate: (1) Maximum velocity achieved, (2) Velocity at t = 3 s, (3) Total displacement at t = 3 s.
Solution Step 1: Derive velocity function v(t):
v(t) = v(0) + Integral_0^t a(tau) dtau = 4 + Integral_0^t (12*tau - 3*tau^2) dtau.
v(t) = 4 + [6*t^2 - t^3] m/s.
Solution Step 2: Maximum Velocity Condition:
Maximum velocity occurs when dv/dt = a(t) = 0 => 12*t - 3*t^2 = 0 => 3*t*(4 - t) = 0 => t = 4.0 s.
v_max = v(4) = 4 + 6*(4)^2 - (4)^3 = 4 + 96 - 64 = 36.0 m/s.
Solution Step 3: Velocity at t = 3.0 s:
v(3) = 4 + 6*(3)^2 - (3)^3 = 4 + 54 - 27 = 31.0 m/s.
Solution Step 4: Derive displacement function s(t) and evaluate at t = 3.0 s:
s(t) = Integral_0^t v(tau) dtau = Integral_0^t (4 + 6*tau^2 - tau^3) dtau = [4*t + 2*t^3 - 0.25*t^4].
s(3) = 4*(3) + 2*(3)^3 - 0.25*(3)^4 = 12 + 54 - 0.25*(81) = 66 - 20.25 = 45.75 m.
Conclusion: Maximum velocity = 36.0 m/s; Velocity at t=3s = 31.0 m/s; Displacement at t=3s = 45.75 m.

[EXAMPLE 2: D’Alembert’s Principle for Connected Incline Bodies]

Problem Context: Block A (mass m_A = 20 kg) rests on a 30-degree inclined plane (coefficient of kinetic friction mu_k = 0.20) and is connected by a light inextensible cord passing over a frictionless pulley to a hanging Block B (mass m_B = 30 kg). Using D’Alembert’s principle, calculate the acceleration a of the system and cable tension T.
Solution Step 1: Establish Kinematic Relations and Dynamic Equilibrium:
Assuming Block B descends vertically with acceleration a, Block A moves UP the incline with identical acceleration a.
For Block A on incline:
Normal reaction: N_A = m_A * g * cos(30) = 20 * 9.81 * 0.8660 = 169.91 N.
Friction force (opposing motion down incline): F_k = mu_k * N_A = 0.20 * 169.91 = 33.98 N.
Downhill gravity component = m_A * g * sin(30) = 20 * 9.81 * 0.50 = 98.10 N.
Reversed Inertia Force = m_A * a = 20 * a (directed down incline).
Dynamic equilibrium along incline: T - 98.10 - 33.98 - 20*a = 0 => T - 20*a = 132.08 N. (Equation 1)
Solution Step 2: Dynamic Equilibrium for Hanging Block B:
Downside gravity force = m_B * g = 30 * 9.81 = 294.30 N.
Cable tension T (upward); Reversed Inertia Force m_B * a (upward):
Dynamic equilibrium vertically: 294.30 - T - 30*a = 0 => T + 30*a = 294.30 N. (Equation 2)
Solution Step 3: Solve Simultaneous Equations for a and T:

Subtracting Equation 1 from Equation 2:
(30*a + 20*a) = 294.30 - 132.08 => 50*a = 162.22 => a = 3.24 m/s^2.
Substitute a into Equation 1: T = 132.08 + 20*(3.24) = 132.08 + 64.88 = 196.96 N.
Conclusion: System acceleration a = 3.24 m/s^2; Cable tension T = 196.96 N.

[EXAMPLE 3: Work-Energy Method for Runaway Construction Cart]

Problem Context: A construction muck cart of mass m = 800 kg travels at v1 = 15 m/s toward an emergency arresting spring buffer of stiffness k = 60 kN/m (60,000 N/m). If the track is horizontal and rolling resistance is neglected, calculate: (1) Maximum compression x_max of the spring buffer in bringing the cart to rest, (2) Maximum deceleration experienced.
Solution Step 1: Work-Energy Equation (Sigma U_1-2 = Delta KE):
Initial state 1 (before contact): v1 = 15 m/s => KE_1 = 0.5 * m * v1^2 = 0.5 * 800 * (15)^2 = 400 * 225 = 90,000 J (90 kJ).
Final state 2 (at maximum spring compression): v2 = 0 m/s => KE_2 = 0 J.
Work done by arresting spring on cart during compression: U_spring = - 0.5 * k * x_max^2 = - 0.5 * 60,000 * x_max^2 = - 30,000 * x_max^2.
Solution Step 2: Equate Work to Change in Kinetic Energy:
- 30,000 * x_max^2 = 0 - 90,000 => 30,000 * x_max^2 = 90,000.
x_max^2 = 90,000 / 30,000 = 3.0 => x_max = sqrt(3.0) = 1.732 m.
Solution Step 3: Compute Maximum Deceleration:
Maximum retarding force occurs at maximum compression: F_max = k * x_max = 60,000 * 1.732 = 103,923 N.
a_max = F_max / m = 103,923 N / 800 kg = 129.90 m/s^2 (~ 13.2 g).
Conclusion: Maximum spring compression x_max = 1.732 m; maximum deceleration = 129.90 m/s^2.

[EXAMPLE 4: Direct Central Impact and Energy Loss]

Problem Context: A pile driver hammer A (mass m_A = 1200 kg) falling vertically hits a stationary concrete pile cap B (mass m_B = 800 kg) with impact velocity v_A1 = 8.0 m/s (v_B1 = 0). The coefficient of restitution is e = 0.40. Calculate: (1) Post-impact velocities v_A2 and v_B2, (2) Total loss of kinetic energy during impact.
Solution Step 1: Conservation of Linear Momentum:
m_A * v_A1 + m_B * v_B1 = m_A * v_A2 + m_B * v_B2.
(1200 * 8.0) + (800 * 0) = 1200 * v_A2 + 800 * v_B2.
9600 = 1200 * v_A2 + 800 * v_B2 => 12 * v_A2 + 8 * v_B2 = 96 => 3 * v_A2 + 2 * v_B2 = 24. (Equation 1)
Solution Step 2: Coefficient of Restitution Equation:
e = (v_B2 - v_A2) / (v_A1 - v_B1) => 0.40 = (v_B2 - v_A2) / (8.0 - 0).
v_B2 - v_A2 = 0.40 * 8.0 = 3.2 m/s => v_B2 = v_A2 + 3.2. (Equation 2)

Solution Step 3: Solve for post-impact velocities:
Substitute Equation 2 into Equation 1: 3 * v_A2 + 2 * (v_A2 + 3.2) = 24.
5 * v_A2 + 6.4 = 24 => 5 * v_A2 = 17.6 => v_A2 = 3.52 m/s.
v_B2 = 3.52 + 3.2 = 6.72 m/s.
Solution Step 4: Kinetic Energy Loss:
Initial KE = 0.5 * 1200 * (8.0)^2 = 600 * 64 = 38,400 J (38.40 kJ).
Final KE = 0.5 * 1200 * (3.52)^2 + 0.5 * 800 * (6.72)^2 = 600 * 12.39 + 400 * 45.16 = 7434.24 + 18063.36 = 25,497.60 J (25.50 kJ).
Energy Loss Delta KE = 38,400 - 25,497.60 = 12,902.40 J (12.90 kJ or 33.6% loss).
Conclusion: Hammer velocity v_A2 = 3.52 m/s; pile cap velocity v_B2 = 6.72 m/s; Energy lost = 12.90 kJ.

>> 5.7 University Review & Examination Questions

* Q1: State D’Alembert’s Principle and explain how it simplifies the solution of dynamic engineering problems.
Answer: D’Alembert’s Principle states that a dynamic system in motion under applied external forces can be brought into formal static equilibrium by adding a fictitious inertia force equal to the product of the mass and acceleration vector acting in the direction opposite to acceleration (F_inertia = - m * a). Thus, Sigma F + (- m * a) = 0. Its primary significance is that all established analytical tools of statics (free-body diagrams, resolving orthogonal forces, moment equilibrium about instantaneous centers) can be directly applied to dynamic systems without solving coupled second-order differential equations.
* Q2: Derive the Work-Energy Theorem for a particle moving along a general curved spatial path.
Answer: Consider a particle of mass m moving along path s under resultant force F. Tangential Newton’s second law states: F_t = m * a_t = m * v * (dv/ds). Multiplying both sides by differential displacement ds: F_t * ds = m * v * dv. Integrating both sides from initial state (s1, v1) to final state (s2, v2): Integral_s1^s2 (F_t ds) = Integral_v1^v2 (m * v dv). The left side is the total external work done U_1-2. The right side evaluates to: [0.5 * m * v^2]_v1^v2 = 0.5 * m * v2^2 - 0.5 * m * v1^2 = Delta KE. Hence, Sigma U_1-2 = Delta KE.

>> 5.8 Key Takeaways and Revision Points

* Variable acceleration requires direct calculus integration: v = Integral a dt, s = Integral v dt.
* Normal acceleration a_n = v^2/rho directs toward curve center; tangential a_t = dv/dt changes speed.
* D’Alembert’s principle adds reversed inertia force (-m*a) to apply static equilibrium equations.
* Work-Energy theorem (Sigma U = Delta KE) eliminates time; Impulse-Momentum (J = Delta p) connects force and time.